

DEPARTMENT OF APPLIED SCIENCES AND HUMANITIES

Program Name:	B.Sc. (H)	AY	2026-27
Course Name:	Electricity and Magnetism	Class / Sem	III Semester
Faculty Name:	Dr S K Jain	Contact	8755169830

Lectures . No	Topic	Reference	Teaching Method(s)
1	Introduction to the course; Coulomb's law and the principle of superposition	T1 Ch.2	C&T
2	Electric field and field lines; properties of field lines; field due to a point charge	T1 Ch.2	C&T, CHART
3	Electric field due to a uniformly charged ring (derivation on the axis)	T1 Ch.2	C&T
4	Electric flux — concept, definition and physical significance	T1 Ch.2	C&T
5	Gauss's law: statement and proof (integral form), derived from Coulomb's law	T1 Ch.2	C&T
6	Gauss's law in differential form, using the divergence theorem	T1 Ch.2	C&T
7	Applications of Gauss's law: infinite line charge; charged cylindrical conductor	T1 Ch.2	C&T, S/P
8	Applications of Gauss's law: infinite sheet of charge; two parallel charged sheets	T1 Ch.2	C&T, S/P
9	Applications of Gauss's law: charged spherical shell; charged hollow sphere	T1 Ch.2	C&T, S/P
10	Applications of Gauss's law: uniformly charged solid sphere; electrostatic energy	T1 Ch.2	C&T, QUIZ
11	Line integral of the electric field; electric potential difference and potential	T1 Ch.2	C&T
12	Conservative nature of the electrostatic field; relation between E and V	T1 Ch.2	C&T

13	Equipotential surfaces; the electric dipole (potential and field)	T1 Ch.2	C&T, CHART
14	Electrostatic potential energy of a system of point charges	T1 Ch.2	C&T
15	Potential and electric field of a uniformly charged wire	T1 Ch.2	C&T
16	Potential and electric field of a uniformly charged disc	T1 Ch.2	C&T
17	Tutorial: problem solving on Units 1 and 2	T1, T2	GD, QUIZ
18	Magnetic field B and the Lorentz force; force between current elements	T1 Ch.5	C&T
19	Magnetic flux; statement of the Biot–Savart law	T1 Ch.5	C&T
20	Biot–Savart law: field due to a straight current-carrying conductor	T1 Ch.5	C&T, S/P
21	Biot–Savart law: field on the axis of, and at the centre of, a circular current loop	T1 Ch.5	C&T, S/P
22	Force on a current-carrying conductor; torque on a current loop	T1 Ch.5	C&T, DEMO
23	Motion of a charged particle in a uniform magnetic field (cyclotron motion)	T1 Ch.5	C&T
24	Ampere's circuital law: integral and differential forms	T1 Ch.5	C&T
25	Application of Ampere's law: magnetic field of a solenoid	T1 Ch.5	C&T, S/P
26	Application of Ampere's law: magnetic field of a toroid	T1 Ch.5	C&T, S/P
27	Magnetism of matter — introduction; Gauss's law of magnetism (integral and differential forms)	T1 Ch.6	C&T
28	Magnetisation current; distinction between bound and free currents	T1 Ch.6	C&T
29	Relative permeability and magnetic susceptibility; classification of magnetic materials	T1 Ch.6, T2	C&T

30	Magnetisation vector M, magnetic intensity H, and the relation $B = \mu_0(H + M)$	T1 Ch.6	C&T
31	Boundary conditions on B and H at an interface between two media	T1 Ch.6	C&T
32	Stored magnetic energy in matter	T1 Ch.6	C&T
33	Magnetic circuits: analogy with electric circuits; reluctance	T2	C&T, CHART
34	B–H curve and the hysteresis loop: retentivity, coercivity and saturation	T1 Ch.6	C&T, CHART
35	Energy loss in hysteresis; domain theory of ferromagnetism	T1 Ch.6	C&T, S/P
36	Tutorial: problem solving on Units 3 and 4	T1, T2	GD, QUIZ
37	Faraday's law of electromagnetic induction: integral and differential forms	T1 Ch.7	C&T
38	Lenz's law; motional EMF	T1 Ch.7	C&T, DEMO
39	Self-inductance: definition and self-inductance of a solenoid	T1 Ch.7	C&T
40	Mutual inductance: definition, reciprocity theorem, and coaxial solenoids	T1 Ch.7	C&T
41	Energy stored in a magnetic field	T1 Ch.7	C&T
42	AC generator; eddy currents and transformers	T1 Ch.7	C&T, S/P
43	Tutorial: problem solving on Unit 5	T1, T2	GD, QUIZ
44	Comprehensive revision of all five units; doubt-clearing session	T1, T2	GD, QUIZ
	Total Lectures:	44	

Teaching Methods: C&T:Chalk & Talk; S/P: -Slides/PPT;Videos; SEM: Seminar; DEMO; CHART; ET/GL: Expert Talk/Guest Lecture; QUIZ; GD: -Group discussion; RTCS: Real time case studies; JAR:-Journal article review; PD:-Poster design; LAB: Laboratory demonstration.

Text / Reference Books:

T1 – David J. Griffiths, *Introduction to Electrodynamics*, Third Edition (Prentice Hall of India)

T2 A Mahajan and A Rangwala, *Electricity and Magnetism*, McGraw Hill India.

